

Maven Final Armageddon – Complete pom.xml practice for custom plugins

1. Maven Build Automation using maven-jar-plugin (complete practice of pom.xml)

```
Command Prompt
D:\maven-jar-pract\target>cd ..
D:\maven-jar-pract>type pom.xml
<?xml version="1.0" encoding="UTF-8" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>maven-jar-pract</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>maven-jar-pract</name>
  <url>http://maven.apache.org</url>
  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.3.0</version>
        <configuration>
          <archive>
            <manifest>
              <mainClass>com.example.App</mainClass>
            </manifest>
          </archive>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>

D:\maven-jar-pract>java -jar target/maven-jar-pract-1.0-SNAPSHOT.jar
Hello World!
```

2. Spotify Docker Maven Plugin (Complete pom.xml practice)

```
Command Prompt
D:\maven-docker-plugin>mvn clean package dockerfile:build
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:maven-docker-plugin >-----
[INFO] Building maven-docker-plugin 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ maven-docker-plugin ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ maven-docker-plugin ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\maven-docker-plugin\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ maven-docker-plugin ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ maven-docker-plugin ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\maven-docker-plugin\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ maven-docker-plugin ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ maven-docker-plugin ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit.JUnit3Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```
Command Prompt
[INFO] contextDirectory: D:\maven-docker-plugin
[INFO] Building Docker context D:\maven-docker-plugin
[INFO] Path(dockerfile): null
[INFO] Path(contextDirectory): D:\maven-docker-plugin
[INFO]
[INFO] Image will be built as spotify-docker-demo:latest
[INFO]
[INFO] Step 1/4 : FROM eclipse-temurin:17
[INFO]
[INFO] Pulling from library/eclipse-temurin
[INFO] Digest: sha256:b04a8c5d46e210873ffd1af6ad5f4d62c69ed3a6736993556eae60bba1373a23
[INFO] Status: Downloaded newer image for eclipse-temurin:17
[INFO] --> b04a8c5d46e2
[INFO] Step 2/4 : WORKDIR /app
[INFO]
[INFO] --> Using cache
[INFO] --> 15d0471ffc80
[INFO] Step 3/4 : COPY target/maven-docker-plugin-1.0-SNAPSHOT.jar app.jar
[INFO]
[INFO] --> 294e19c226ab
[INFO] Step 4/4 : CMD ["java", "-jar", "app.jar"]
[INFO]
[INFO] --> Running in 8265ab7413a2
[INFO] --> Removed intermediate container 8265ab7413a2
[INFO] --> 7ec19fd9f79b
[INFO] Successfully built 7ec19fd9f79b
[INFO] Successfully tagged spotify-docker-demo:latest
[INFO]
[INFO] Detected build of image with id 7ec19fd9f79b
[INFO] Building jar: D:\maven-docker-plugin\target\maven-docker-plugin-1.0-SNAPSHOT-docker-info.jar
[INFO] Successfully built spotify-docker-demo:latest
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 46.214 s
[INFO] Finished at: 2026-05-13T23:14:38+05:30
[INFO]
D:\maven-docker-plugin>docker run spotify-docker-demo:latest
Hello World!

D:\maven-docker-plugin>
```

```
Command Prompt
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>maven-docker-plugin</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>maven-docker-plugin</name>
  <url>http://maven.apache.org</url>
  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.3.0</version>
        <configuration>
          <archive>
            <manifest>
              <mainClass>com.example.App</mainClass>
            </manifest>
          </archive>
        </configuration>
      </plugin>
      <plugin>
        <groupId>com.spotify</groupId>
        <artifactId>dockerfile-maven-plugin</artifactId>
        <version>1.4.13</version>
        <configuration>
          <repository>spotify-docker-demo</repository>
          <tag>latest</tag>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

3. Containerizing a Java Application Using Jib and Maven

```
Command Prompt
D:\googlejib-pract>notepad pom.xml

D:\googlejib-pract>mvn clean package jib:dockerBuild
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/maven-metadata.xml (708 B at 624 B/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/jib-maven-plugin/3.4.3/jib-maven-plugin-3.4.3.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/jib-maven-plugin/3.4.3/jib-maven-plugin-3.4.3.pom (4.9 kB at 35 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.15.2/jackson-bom-2.15.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-bom/2.15.2/jackson-bom-2.15.2.pom (18 kB at 108 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.15/jackson-parent-2.15.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jackson/jackson-parent/2.15/jackson-parent-2.15.pom (6.5 kB at 57 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/jib-maven-plugin/3.4.3/jib-maven-plugin-3.4.3.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/google/cloud/tools/jib-maven-plugin/3.4.3/jib-maven-plugin-3.4.3.jar (566 kB at 1.6 MB/s)
[INFO]
[INFO] -----< com.example:googlejib-pract >-----
[INFO] Building googlejib-pract 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ googlejib-pract ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ googlejib-pract ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\googlejib-pract\src\main\resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ googlejib-pract ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding windows-1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ googlejib-pract ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\googlejib-pract\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ googlejib-pract ---
```

```
credentials not found in native keychain

[WARNING] The credential helper (docker-credential-desktop) has nothing for server URL: registry.hub.docker.com
[WARNING]
Got output:

credentials not found in native keychain

[INFO] Using credentials from Docker config (C:\Users\sreev\.docker\config.json) for eclipse-temurin:17-jre
[INFO] Using base image with digest: sha256:0d79988c68791ce864fe39d149ab1dc84f680539dca77ee7f6f3b041ad7f2f43
[INFO]
[INFO] Container entrypoint set to [java, -cp, @/app/jib-classpath-file, com.example.App]
[INFO]
[INFO] Built image to Docker daemon as jib-demo
[INFO] Executing tasks:
[INFO] [=====] 100.0% complete
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 39.983 s
[INFO] Finished at: 2026-05-13T23:37:36+05:30
[INFO] -----

D:\googlejib-pract>docker run jib-demo
Hello World!

D:\googlejib-pract>
```

```

Command Prompt
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>googlejib-pract</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>googlejib-pract</name>
<url>http://maven.apache.org</url>
<properties>
  <maven.compiler.source>17</maven.compiler.source>
  <maven.compiler.target>17</maven.compiler.target>
</properties>
<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-jar-plugin</artifactId>
      <version>3.3.0</version>
      <configuration>
        <archive>
          <manifest>
            <mainClass>com.example.App</mainClass>
          </manifest>
        </archive>
      </configuration>
    </plugin>
    <plugin>
      <groupId>com.google.cloud.tools</groupId>
      <artifactId>jib-maven-plugin</artifactId>
      <version>3.3.2</version>
      <configuration>
        <to>
          <image>jib-demo</image>
        </to>
      </configuration>
    </plugin>
  </plugins>
</build>
<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>3.8.1</version>
    <scope>test</scope>
  </dependency>
</dependencies>
</project>

```

4. Deploying a Maven Web Application Using Tomcat7 Maven Plugin

```

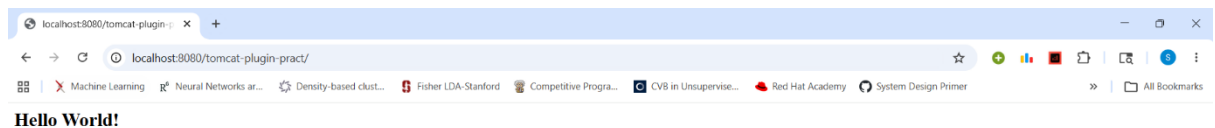
Command Prompt
D:\tomcat-plugin-pract>type pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>tomcat-plugin-pract</artifactId>
  <packaging>war</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>tomcat-plugin-pract Maven Webapp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <finalName>tomcat-plugin-pract</finalName>
    <plugins>
      <plugin>
        <groupId>org.apache.tomcat.maven</groupId>
        <artifactId>tomcat7-maven-plugin</artifactId>
        <version>2.2</version>
      </plugin>
    </plugins>
  </build>
</project>

```

```

D:\tomcat-plugin-pract>mvn tomcat7:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:tomcat-plugin-pract >-----
[INFO] Building tomcat-plugin-pract Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] >>> tomcat7:2.2:run (default-cli) > process-classes @ tomcat-plugin-pract >>>
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ tomcat-plugin-pract ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ tomcat-plugin-pract ---
[INFO] No sources to compile
[INFO]
[INFO] <<< tomcat7:2.2:run (default-cli) < process-classes @ tomcat-plugin-pract <<<
[INFO]
[INFO] --- tomcat7:2.2:run (default-cli) @ tomcat-plugin-pract ---
[INFO] Running war on http://localhost:8080/tomcat-plugin-pract
[INFO] Using existing Tomcat server configuration at D:\tomcat-plugin-pract\target\tomcat
[INFO] create webapp with contextPath: /tomcat-plugin-pract
May 14, 2026 10:28:07 AM org.apache.coyote.AbstractProtocol init
INFO: Initializing ProtocolHandler ["http-bio-8080"]
May 14, 2026 10:28:07 AM org.apache.catalina.core.StandardService startInternal
INFO: Starting service Tomcat
May 14, 2026 10:28:07 AM org.apache.catalina.core.StandardEngine startInternal
INFO: Starting Servlet Engine: Apache Tomcat/7.0.47
May 14, 2026 10:28:08 AM org.apache.coyote.AbstractProtocol start
INFO: Starting ProtocolHandler ["http-bio-8080"]

```



5. Deploying a Maven Web Application Using Jetty Maven Plugin

```

Command Prompt - mvn jett
D:\jetty-maven-pract>mvn jetty:run
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:jetty-maven-pract >-----
[INFO] Building jetty-maven-pract Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] >>> jetty:11.0.20:run (default-cli) > test-compile @ jetty-maven-pract >>>
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ jetty-maven-pract ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ jetty-maven-pract ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ jetty-maven-pract ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory D:\jetty-maven-pract\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ jetty-maven-pract ---
[INFO] No sources to compile
[INFO]
[INFO] <<< jetty:11.0.20:run (default-cli) < test-compile @ jetty-maven-pract <<<
[INFO]
[INFO] --- jetty:11.0.20:run (default-cli) @ jetty-maven-pract ---
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/jakarta/transaction/jakarta.transaction-api/2.0.0/jakarta.transaction-api-2.0.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/jakarta/transaction/jakarta.transaction-api/2.0.0/jakarta.transaction-api-2.0.0.pom (12 kB at 12 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools-api/3.11.0/maven-plugin-tools-api-3.11.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools-api/3.11.0/maven-plugin-tools-api-3.11.0.pom (6.0 kB at 76 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools/3.11.0/maven-plugin-tools-3.11.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugin-tools/maven-plugin-tools/3.11.0/maven-plugin-tools-3.11.0.pom (17 kB at 154 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/2.0.9/slf4j-api-2.0.9.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-api/2.0.9/slf4j-api-2.0.9.pom (2.8 kB at 36 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-parent/2.0.9/slf4j-parent-2.0.9.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-parent/2.0.9/slf4j-parent-2.0.9.pom (16 kB at 180 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-bom/2.0.9/slf4j-bom-2.0.9.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/slf4j/slf4j-bom/2.0.9/slf4j-bom-2.0.9.pom (4.9 kB at 78 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/apache-jsp/11.0.20/apache-jsp-11.0.20.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/apache-jsp/11.0.20/apache-jsp-11.0.20.pom (3.5 kB at 21 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-util/11.0.20/jetty-util-11.0.20.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/jetty-util/11.0.20/jetty-util-11.0.20.pom (3.7 kB at 22 kB/s)

```

```

(44 kB at 12 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/jetty/websocket/websocket-jetty-api/11.0.20/websocket-jetty-api-11.0.20.jar (41 kB at 10 kB/s)
[INFO] Configuring Jetty for project: jetty-maven-pract Maven Webapp
[INFO] Classes = D:\jetty-maven-pract\target\classes
[INFO] Context path = /
[INFO] Tmp directory = D:\jetty-maven-pract\target\tmp
[INFO] web.xml file = file:///D:/jetty-maven-pract/src/main/webapp/WEB-INF/web.xml
[INFO] Webapp directory = D:\jetty-maven-pract\src\main\webapp
[INFO] Web defaults = org/eclipse/jetty/webapp/webdefault.xml
[INFO] Web overrides = none
[INFO] jetty-11.0.20; built: 2024-01-29T21:04:22.394Z; git: 922f8dc188f7011e60d0361de585fd4ac4d63064; jvm 17.0.18+8
[INFO] Session workerName=node0
[INFO] Started o.e.j.m.p.MavenWebAppContext@1ab53860{Archetype Created Web Application,/, [file:///D:/jetty-maven-pract/src/main/webapp/], AVAILABLE}{file:///D:/jetty-maven-pract/src/main/webapp/}
[INFO] Started ServerConnector@382d549a{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
[INFO] Started Server@113dcaf8{STARTING}[11.0.20,sto=0] @42633ms
[INFO] Automatic redeployment disabled, see 'mvn jetty:help' for more redeployment options

```

localhost:8080

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Machine Learning Neural Networks ar... Density-based clust... Fisher LDA-Stanford Competitive Prog... CVB in Unsupervise... Red Hat Academy System Design Primer

» All Bookmarks

Hello World!

```
D:\jetty-maven-pract>type pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>jetty-maven-pract</artifactId>
  <packaging>war</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>jetty-maven-pract Maven Webapp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <finalName>jetty-maven-pract</finalName>
    <plugins>
      <plugin>
        <groupId>org.eclipse.jetty</groupId>
        <artifactId>jetty-maven-plugin</artifactId>
        <version>11.0.20</version>
      </plugin>
    </plugins>
  </build>
</project>
```